



ALBERT PAMONAG ENGINEERING CONSULTANCY

DESIGN SHEET

Project : Project name

Member : F-1 - Grid A-1

Designed / Date : APEC

03-Oct-2026

JACKETING (ENLARGEMENT) OF AN EXISTING ISOLATED FOOTING

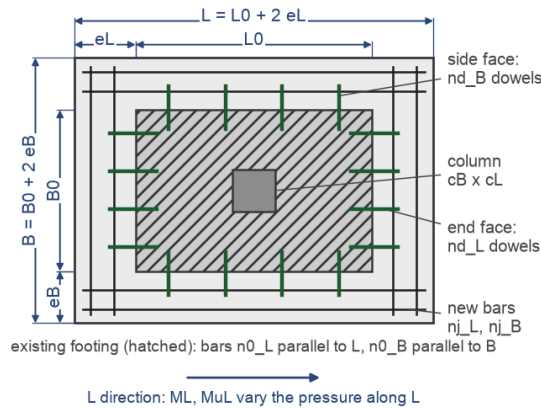
ACI 562-25 - Assessment, Repair and Rehabilitation of Existing Concrete Structures; ACI 318-25M; ASCE 41-17

Yellow cells are inputs. Every other cell is locked. Owner: Albert Pamonag - Albert Pamonag Engineering Consultancy.

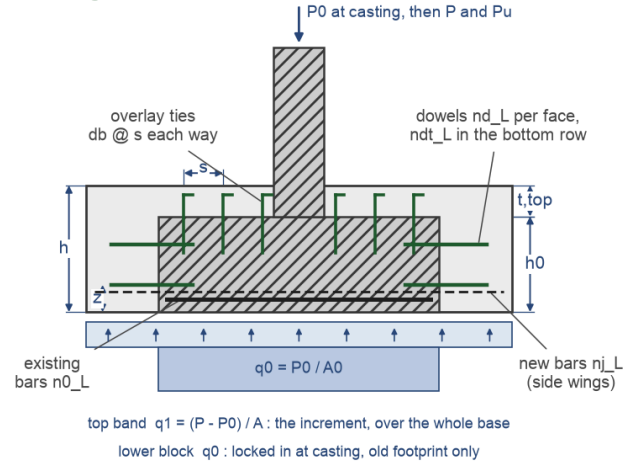
Description	Symbol	Value	Unit	Equation / reference	Status
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1.0 INPUT DATA

Plan



Section along L



Input reference only - not to scale, not linked to the inputs

Existing footing

Existing footing width	B0	2000.0	mm	
Existing footing length	L0	3000.0	mm	
Existing footing thickness	h0	600.0	mm	
Existing concrete strength	f'c0	21.0	MPa	
Existing reinforcement yield	fy0	275	MPa	
Existing bottom bars parallel to L	n	8		outer (lower) layer, across the width B0
Their diameter	db	20.0	mm	
Existing bottom bars parallel to B	n	15		inner layer, across the length L0
Their diameter	db	20.0	mm	
Clear cover to the existing outer layer	cc0	75.0	mm	

Column

Column dimension parallel to B	cB	400.0	mm	at the top of the footing (jacketed size if the column is jacketed)
Column dimension parallel to L	cL	400.0	mm	
Column location factor	alpha_s	40		ACI 318-25M 22.6.5.3: 40 interior, 30 edge, 20 corner
Overlay tied to the column?		Yes		No: shear uses the existing depth only (see 5.0, 6.0)

Jacket

Wing added on EACH side, B direction	eB	600.0	mm	0 = no enlargement in this direction
Wing added on EACH side, L direction	eL	600.0	mm	
Bonded overlay thickness on top	t_top	200.0	mm	0 = none; the wings are cast to the full depth h0 + t_top
Jacket concrete strength	f'cj	28.0	MPa	
Jacket reinforcement yield	fyj	415	MPa	
New bottom bars parallel to L	n	8		total in the two side wings, continuous over L
Their diameter	db	20.0	mm	
New bottom bars parallel to B	n	10		total in the two end wings, continuous over B
Their diameter	db	20.0	mm	
Clear cover to the new outer layer	ccj	75.0	mm	
Lightweight concrete factor	lambda	1.00		

Interfaces

Surface of the old concrete		roughened		roughened = ~6 mm amplitude, ACI 318-25M 26.5.6.2(e); ACI 562-25 9.4.4.3
Dowels crossing EACH end face	n	14		end faces: the old faces of width B0
Of these, in the bottom (tension) row	n	7		
Dowels crossing EACH side face	n	20		side faces: the old faces of length L0
Of these, in the bottom (tension) row	n	10		
Dowel diameter	db	20.0	mm	
Dowel yield	fy	415	MPa	

Description	Symbol	Value	Unit	Equation / reference	Status
Bottom dowel row above the soffit	z	100.0	mm		
Overlay tie diameter	db	12.0	mm	0 = no ties	
Overlay tie spacing, each way	s	300.0	mm	square grid	
Overlay tie yield	fyt	415	MPa		

Construction sequence and loads (at the top of the footing)

Load relieved before the jacket is cast?		No		ACI 562-25 R9.4.1; ASCE 41-17 C8.7 - governs the design	
Service load on the footing at casting	P0	700.0	kN		
Service moment at casting, along L	M0L	0.0	kN-m		
Service moment at casting, along B	M0B	0.0	kN-m		
Final service load	P	1500.0	kN		
Final service moment, along L	ML	180.0	kN-m	varies the pressure along L	
Final service moment, along B	MB	60.0	kN-m		
Permissible net bearing pressure	qa	200.0	kPa	for this service combination ACI 318-25M 13.3.1.1	
Factored load	Pu	2000.0	kN		
Factored moment, along L	MuL	260.0	kN-m		
Factored moment, along B	MuB	90.0	kN-m		

Input check every cell a number; column inside the footing; steel at the face **OK**

Loads are those at the top of the footing; qa is the net permissible pressure. Enter moments as magnitudes. Sample values: a 2.0 x 3.0 x 0.60 m footing enlarged by 600 mm on every side with a 200 mm overlay, 700 kN on it when the jacket is cast.

Background on footing strengthening and why it is rarely straightforward: Antoniou (2023), Section 4.18. The equations behind each section, with clause and page, are on the Theory sheet.

RESULT SUMMARY

Bearing: q,max <= qa	q,max / qa	0.975	ACI 318-25M 13.3.1.1 with the load history	PASS
Whole base in contact	q,min	32.0 kPa	least of the service and factored q,min; needs >= 0	PASS
Flexure at the column face	Mu / phi Mn	0.794	ACI 318-25M Table 13.2.7.1, 22.3.3.2	PASS
Minimum flexural steel (face, wing root)	As,min / As	0.978	ACI 318-25M 8.6.1.1, 7.6.1.1; d >= 150 mm (13.3.1.2)	PASS
Minimum steel at the column for punching	req / prov	0.952	ACI 318-25M Eq. (8.6.1.2)	PASS
One-way shear at d from the column face	Vu / phi Vc	0.433	ACI 318-25M 13.2.7.2, 13.2.6.2(a)	PASS
Two-way shear with moment transfer	vu / phi vc	0.590	ACI 318-25M Table 22.6.5.2, 8.4.4.2	PASS
Wings: flexure and shear at the old faces	D/C	0.175	cantilever from the old footing face	PASS
Shear friction on the old faces, surface	Vu / phi Vn	0.169	ACI 318-25M Eq. (22.9.4.2); ACI 562-25 9.4.4.3	PASS
Overlay: horizontal shear and tie spacing	Vu / phi Vnh	0.321	ACI 318-25M 16.4.3.1, 16.4.7.2	PASS
OVERALL - FOOTING JACKET				PASS

Unshored - the load on the footing when the jacket is cast stays on the old footprint and only the increment spreads over the enlarged base (ACI 562-25 R9.4.1; ASCE 41-17 C8.7). The old footprint carries q0 + q1, the wings q1 alone.

2.0 GEOMETRY AND MATERIALS

Jacketed width	B	3200.0 mm	B0 + 2 eB	
Jacketed length	L	4200.0 mm	L0 + 2 eL	
Jacketed thickness	h	800.0 mm	h0 + t,top	
Existing base area	A0	6.000 m2		
Jacketed base area	A	13.440 m2		
Concrete strength used for every section	f'c	21.00 MPa	the lesser f'c ACI 318-25M 22.3.3.4, 22.5.4.3, 22.9.4.4	
Square root of f'c used in shear		4.583 MPa	not more than 8.3 MPa ACI 318-25M 22.5.3.1, 22.6.3.1	
Stress block factor	beta1	0.850	ACI 318-25M Table 22.2.2.4.3	
Existing steel parallel to L	As0	2,513.3 mm2		
Existing steel parallel to B	As0	4,712.4 mm2		
New steel parallel to L (side wings)	Asj	2,513.3 mm2		
New steel parallel to B (end wings)	Asj	3,141.6 mm2		
Bottom-row dowels, each end face	Asd	2,199.1 mm2		
Bottom-row dowels, each side face	Asd	3,141.6 mm2		
Depth to the existing bars parallel to L	d0	715.0 mm	from the top of the jacketed footing	
Depth to the existing bars parallel to B	d0	695.0 mm	inner layer	
Depth to the new bars parallel to L	dj	715.0 mm		
Depth to the new bars parallel to B	dj	695.0 mm		
Depth to the bottom dowel row	dd	700.0 mm		
Wings provided?		Yes		
Overlay provided?		Yes		
Full depth counted in shear?		Yes	No: an overlay not tied to the column is left out	

Directions: 'L' rows are the cantilevers that project along L - their sections are B wide and their bars run parallel to L. 'B' rows are the same turned through 90 degrees.

3.0 LOAD HISTORY AND BEARING (ACI 562-25 R9.4.1, ASCE 41-17 C8.7)

Load locked on the old footprint	P0	700.0 kN	0 if the load is relieved before casting	
Moment locked on the old footprint, L	M0L	0.0 kN-m		
Moment locked on the old footprint, B	M0B	0.0 kN-m		
Service increment on the jacketed base	dP	800.0 kN	P - P0	
Service moment increment, L	dML	180.0 kN-m		
Service moment increment, B	dMB	60.0 kN-m		
Factored increment on the jacketed base	dPu	1,300.0 kN	Pu - P0: the locked-in load is not factored	
Factored moment increment, L	dMuL	260.0 kN-m		
Factored moment increment, B	dMuB	90.0 kN-m		

Description	Symbol	Value	Unit	Equation / reference	Status
Pressure locked under the old footprint	q0	116.67	kPa	P0 / A0	
Pressure from the increment, whole base	q1	59.52	kPa	(P - P0) / A	
Moment term at the old-footprint corner		18.90	kPa	6 M0/(B0 L0^2) + 6 dM L0/(B L^3), both directions	
Moment term at the wing corner		27.50	kPa	6 dM/(B L^2), both directions	
Greatest pressure under the old footprint	q,old	195.09	kPa	q0 + q1 + moment term	
Greatest pressure under the wings	q,wing	87.03	kPa	q1 + moment term	
Greatest service pressure	q,max	195.09	kPa		
Bearing utilisation	q,max / qa	0.975		ACI 318-25M 13.3.1.1	PASS
Least service pressure	q,min	32.02	kPa	q,min >= 0: whole base in contact	PASS
Factored pressure from the increment	qu1	96.73	kPa	(Pu - P0) / A	
Factored moment term, old corner		27.59	kPa		
Factored moment term, wing corner		40.19	kPa		
Greatest factored pressure, old footprint	qu,old	240.98	kPa		
Greatest factored pressure, wings	qu,wing	136.92	kPa		
Least factored pressure	qu,min	56.53	kPa	qu,min >= 0: whole base in contact	PASS

For an upper bound on the jacket demands (the existing pressure fully redistributed at ultimate), run the sheet with the existing load relieved; the bearing check is then the lower bound.

4.0 FLEXURE AT THE COLUMN FACE (ACI 318-25M Table 13.2.7.1, 22.3.3)

L direction - section B wide, bars parallel to L

Old cantilever beyond the column face	a0	1300.0	mm		
Jacketed cantilever beyond the face	a	1900.0	mm		
Factored moment at the column face	Mu	867.3	kN-m	locked-in pressure on a0 + increment on a, with moments	
Tension steel crossing the section	As	5,026.5	mm2	existing bars + new bars in the side wings	
Tension force at yield	T	1,734.2	kN	As0 fy0 + Asj fyj	
Depth of the stress block	a	30.36	mm	T / (0.85 fc b), whole composite width	22.3.3.2
Neutral axis depth	c	35.72	mm		
Effective depth (centroid of the steel)	d	715.0	mm		
Net tensile strain	eps_t	0.05705		0.003 (dt - c) / c	
Strength reduction factor	phi	0.900		ACI 562-25 Table 5.3.2: 0.90 if eps_t >= eps_ty + 0.003	
Every tension layer yields?		Yes		required by the yield-stress block used here	
Nominal moment strength	Mn	1,213.6	kN-m	sum of As fy (d - a/2)	
Design moment strength	phi Mn	1,092.2	kN-m		
Utilisation	Mu / phi Mn	0.794			PASS
Longitudinal force of one adjoining wing	N	0.0	kN	Asj fyj / 2 - 0.85 fc a e x Mu/Mn; 0 with an overlay	see 8.0
Minimum flexural reinforcement	As,min	4,608.0	mm2	0.0018 Ag, jacketed section	ACI 318-25M 8.6.1.1, 7.6.1.1
As,min / As		0.917			PASS

B direction - section L wide, bars parallel to B

Old cantilever beyond the column face	a0	800.0	mm		
Jacketed cantilever beyond the face	a	1400.0	mm		
Factored moment at the column face	Mu	546.7	kN-m	locked-in pressure on a0 + increment on a, with moments	
Tension steel crossing the section	As	7,854.0	mm2	existing bars + new bars in the end wings	
Tension force at yield	T	2,599.7	kN	As0 fy0 + Asj fyj	
Depth of the stress block	a	34.68	mm	T / (0.85 fc b), whole composite width	22.3.3.2
Neutral axis depth	c	40.80	mm		
Effective depth (centroid of the steel)	d	695.0	mm		
Net tensile strain	eps_t	0.04811		0.003 (dt - c) / c	
Strength reduction factor	phi	0.900		ACI 562-25 Table 5.3.2: 0.90 if eps_t >= eps_ty + 0.003	
Every tension layer yields?		Yes		required by the yield-stress block used here	
Nominal moment strength	Mn	1,761.7	kN-m	sum of As fy (d - a/2)	
Design moment strength	phi Mn	1,585.5	kN-m		
Utilisation	Mu / phi Mn	0.345			PASS
Longitudinal force of one adjoining wing	N	0.0	kN	Asj fyj / 2 - 0.85 fc a e x Mu/Mn; 0 with an overlay	see 8.0
Minimum flexural reinforcement	As,min	6,048.0	mm2	0.0018 Ag, jacketed section	ACI 318-25M 8.6.1.1, 7.6.1.1
As,min / As		0.770			PASS
Least effective depth of the bottom steel	d	695.0	mm	at least 150 mm	ACI 318-25M 13.3.1.2
					PASS

Mn uses the entire composite section and makes no distinction between shored and unshored construction (ACI 318-25M 22.3.3.2, 22.3.3.3). That needs the interfaces of 8.0 and 9.0 to transfer their forces.

5.0 ONE-WAY SHEAR (ACI 318-25M 13.2.7.2, 13.2.6.2(a))

L direction

Depth used for shear	dv	715.0	mm	less the overlay unless it is tied to the column	
Critical section, from the footing centre		915.0	mm	dv from the column face	ACI 318-25M 7.4.3.2
Old footprint beyond the section		585.0	mm		
Jacketed footing beyond the section		1185.0	mm		
Factored shear at the section	Vu	578.5	kN	pressure beyond the section, both stages	
Design shear strength	phi Vc	1,336.8	kN	phi 0.17 lambda sqrt(fc) bw dv, phi = 0.75	13.2.6.2(a), 22.5.4.4
Utilisation	Vu / phi Vc	0.433			PASS

B direction

Depth used for shear	dv	695.0	mm	less the overlay unless it is tied to the column	
Critical section, from the footing centre		895.0	mm	dv from the column face	ACI 318-25M 7.4.3.2

Description	Symbol	Value	Unit	Equation / reference	Status
Old footprint beyond the section		105.0	mm		
Jacketed footing beyond the section		705.0	mm		
Factored shear at the section	Vu	352.1	kN	pressure beyond the section, both stages	
Design shear strength	phi Vc	1,705.5	kN	phi 0.17 lambda sqrt(f'c) bw dv, phi = 0.75 13.2.6.2(a), 22.5.4.4	
Utilisation	Vu / phi Vc	0.206			PASS

ACI 318-25M 13.2.6.2 applies to a footing designed as rigid, with the linear pressure used here; phi = 0.75 is ACI 562-25 Table 5.3.2. The d offset of ACI 318-25M 7.4.3.2 rests on the column reaction, which enters at the top of the OLD footing unless the overlay is tied to the column.

6.0 TWO-WAY SHEAR (ACI 318-25M 22.6, 8.4.4.2)

Average effective depth	d	705.0	mm	ACI 318-25M 22.6.2.1	
Depth used for punching	dp	705.0	mm	less the overlay unless it is tied to the column	
Critical section side parallel to L	b1	1105.0	mm	d/2 from the column faces ACI 318-25M 22.6.4.1	
Critical section side parallel to B	b2	1105.0	mm		
Critical perimeter	bo	4420.0	mm		
Factored shear on the perimeter	Vu	1,739.4	kN	Pu less the pressure inside the perimeter, both stages	
Shear stress from Vu	vuv	0.5582	MPa	Vu / (bo d)	
Fraction of MuL transferred by shear	gamma_v	0.400		ACI 318-25M Eq. (8.4.2.2.2), Eq. (8.4.4.2.2)	
Fraction of MuB transferred by shear	gamma_v	0.400			
Polar property of the section, for MuL	Jc	6.987E+11	mm4	interior column ACI 318-25M R8.4.4.2.3	
Polar property of the section, for MuB	Jc	6.987E+11	mm4		
Shear stress from MuL		0.0822	MPa	gamma_v Msc c / Jc, Msc = the column moment	
Shear stress from MuB		0.0285	MPa		
Greatest factored shear stress	vu	0.6689	MPa	both moments added at the corner of the section	
Column aspect ratio	beta	1.000			
(a) 0.33 lambda sqrt(f'c)		1.5122	MPa	ACI 318-25M Table 22.6.5.2	
(b) (0.17 + 0.33/beta) lambda sqrt(f'c)		2.2913	MPa		
(c) with 0.17 + 0.083 alpha_s d / bo		3.2057	MPa	(0.17 + 0.083 alpha_s d/bo) lambda sqrt(f'c)	
Two-way shear strength	vc	1.5122	MPa	least of (a), (b), (c); lambda_s = 1.0 by 13.2.6.2(b)	
Design two-way shear strength	phi vc	1.1342	MPa		
Utilisation	vu / phi vc	0.590			PASS

An overlay cast around a column that is not tied to it does not move the loaded area up: the column still bears on the old footing, so the punching depth is the existing one. Tie the overlay to the column (column jacket, dowels) to count it. ACI 318-25M 22.6 has no composite-member clause; the lesser f'c is used.

Minimum reinforcement at the column - ACI 318-25M Eq. (8.6.1.2)

Size effect factor	lambda_s	0.7236		ACI 318-25M Eq. (22.5.5.1.3)	
Shear stress above which it applies		0.4228	MPa	phi 0.17 sqrt(f'c) lambda_s lambda_b	
Eq. (8.6.1.2) applies?	Yes			vuv > the stress above	
Thickness counted at the column	h	800.0	mm		
Width b,slab for the bars parallel to L	b,slab	2800.0	mm	column + 1.5 h each side ACI 318-25M 8.4.2.2.3	
As fy of those bars within b,slab		1,386.5	kN	bars taken as evenly spread over their zones	
As,min fy required within b,slab		1,151.4	kN	5 vuv b,slab bo / (phi alpha_s)	
Required / provided		0.830			PASS
Width b,slab for the bars parallel to B	b,slab	2800.0	mm	column + 1.5 h each side ACI 318-25M 8.4.2.2.3	
As fy of those bars within b,slab		1,209.5	kN	bars taken as evenly spread over their zones	
As,min fy required within b,slab		1,151.4	kN	5 vuv b,slab bo / (phi alpha_s)	
Required / provided		0.952			PASS

A lightly reinforced footing punches when the bars near the column yield, below the Table 22.6.5.2 strength. ACI 318-25M 13.2.6.2(b) lifts the size effect from the strength, not from this trigger. Existing bars cannot be added to: if this fails, lower vuv (larger column, or an overlay tied to it).

7.0 WINGS - CANTILEVER FROM THE OLD FOOTING FACE

L direction - wings beyond the end faces

Wing in this direction?	Yes				
Factored shear at the old face	Vu,w	231.2	kN	increment only; corners counted in both directions	
Factored moment at the old face	Mu,w	70.1	kN-m		
Tension steel crossing the old face	As,w	4,712.4	mm2	bottom-row dowels + the continuous new bars	
Depth of the stress block	a	34.24	mm		
Effective depth	d,w	708.0	mm		
Strength reduction factor	phi	0.900		ACI 562-25 Table 5.3.2	
Every tension layer yields?	Yes				
Design moment strength	phi Mn,w	1,216.0	kN-m		
Flexure utilisation	Mu / phi Mn	0.058			PASS
As,min / As,w at the old face		0.978		0.0018 Ag ACI 318-25M 7.6.1.1, 8.6.1.1	PASS
Design shear strength of the wing	phi Vc,w	1,323.7	kN	ACI 318-25M 13.2.6.2(a), at the face	
Shear utilisation	Vu / phi Vc	0.175			PASS

B direction - wings beyond the side faces

Wing in this direction?	Yes				
Factored shear at the old face	Vu,w	269.5	kN	increment only; corners counted in both directions	
Factored moment at the old face	Mu,w	81.4	kN-m		
Tension steel crossing the old face	As,w	6,283.2	mm2	bottom-row dowels + the continuous new bars	
Depth of the stress block	a	34.78	mm		
Effective depth	d,w	697.5	mm		
Strength reduction factor	phi	0.900		ACI 562-25 Table 5.3.2	

Description	Symbol	Value	Unit	Equation / reference	Status
Every tension layer yields?		Yes			
Design moment strength	ϕM_n , w	1,596.1	kN-m		
Flexure utilisation	μ / ϕM_n	0.051			PASS
A_s , min / A_s , w at the old face		0.963		0.0018 Ag ACI 318-25M 7.6.1.1, 8.6.1.1	PASS
Design shear strength of the wing	ϕV_c , w	1,711.6	kN	ACI 318-25M 13.2.6.2(a), at the face	
Shear utilisation	V_u / ϕV_c	0.157			PASS

The wing shear is taken at the old face, not at d from it: the reaction there is a shear across a joint and puts no compression into the end of the wing (ACI 318-25M 7.4.3.2). The existing bars stop at the old face: its tension is carried by the bottom dowels and the new bars alone (ACI 562-25 9.4.5.2), and only if the dowels are developed on both sides - see 10.0.

8.0 SHEAR FRICTION ON THE OLD FACES (ACI 318-25M 22.9.4)

Surface preparation				ACI 562-25 9.4.4.3: an intentionally roughened substrate	PASS
Coefficient of friction	μ	1.00		ACI 318-25M Table 22.9.4.2: (b) 1.0 λ , (c) 0.6	
Yield used for shear friction	f_y	415	MPa	$f_y \leq 420$ MPa ACI 318-25M Table 20.2.2.4(a)	
Upper limit on V_n / A_c		4.20	MPa	ACI 318-25M Table 22.9.4.4, lesser f_c	

Each end face - carries the L-direction wing

Factored shear on the face	V_u	231.2	kN	resultant of the wing shear V_u , w and 2 N along the face	
Dowels crossing the face	A_{vf}	4,398.2	mm ²		
Area of the face	A_c	1,200,000	mm ²	old footing face	
Nominal strength, uncapped	V_n	1,825.3	kN	$\mu A_{vf} f_y$ ACI 318-25M Eq. (22.9.4.2), $N_u = 0$	
Upper limit	$V_{n,max}$	5,040.0	kN		
Nominal strength	V_n	1,825.3	kN		
Design strength	ϕV_n	1,368.9	kN	$\phi = 0.75$ ACI 562-25 Table 5.3.2	
Utilisation	V_u / ϕV_n	0.169			PASS
Dowel area required	$A_{vf,req}$	742.8	mm ²	V_u / ($\phi \mu f_y$)	

Each side face - carries the B-direction wing

Factored shear on the face	V_u	269.5	kN	resultant of the wing shear V_u , w and 2 N along the face	
Dowels crossing the face	A_{vf}	6,283.2	mm ²		
Area of the face	A_c	1,800,000	mm ²	old footing face	
Nominal strength, uncapped	V_n	2,607.5	kN	$\mu A_{vf} f_y$ ACI 318-25M Eq. (22.9.4.2), $N_u = 0$	
Upper limit	$V_{n,max}$	7,560.0	kN		
Nominal strength	V_n	2,607.5	kN		
Design strength	ϕV_n	1,955.6	kN	$\phi = 0.75$ ACI 562-25 Table 5.3.2	
Utilisation	V_u / ϕV_n	0.138			PASS
Dowel area required	$A_{vf,req}$	865.7	mm ²	V_u / ($\phi \mu f_y$)	

ACI 562-25 9.3.3 and 9.4.4 send the interface to the design-basis code. Dowels in the flexural tension zone serve both the moment and the shear friction (ACI 318-25M R22.9.4.5). Any shear carried by the overlay above the old footing is neglected.

N is the part of a wing's bar force that its own share of the stress block does not balance (4.0). Without an overlay it crosses the old face beside that wing as a shear along the face - each half of the face takes half of V_u , w and one N , resolved into one shear (ACI 318-25M 22.3.3.1; ACI 562-25 R9.4.5.3). With an overlay the transfer is the horizontal one of 9.0. An unroughened surface is calculated for assessment only.

9.0 OVERLAY - HORIZONTAL SHEAR (ACI 318-25M 16.4)

Ties across the contact surface?		Yes			
Tie area per unit contact area	$A_{vf}/(b_v s)$	0.001257			
Tie yield used	f_{yt}	415	MPa	$f_{yt} \leq 420$ MPa ACI 318-25M Table 20.2.2.4(a)	
Minimum		0.000843		$A_{vf,min} = 0.35 b_v s / f_y$ ACI 318-25M 16.4.6.1	
Ties meet the minimum?		Yes		needed for rows (b), (c) and (d)	
Transfer by bond alone permitted?		Yes		ACI 318-25M 16.4.1.2: wings can lift the overlay	
(a) cementitious bond		0.550	MPa	0.55 $b_v d$ ACI 318-25M Table 16.4.4.1	
(c) roughened, with minimum ties		2.113	MPa	$\lambda (1.8 + 0.6 A_{vf} f_{yt}/(b_v s)) \leq 3.5$	
(b), (d) shear friction over d		0.522	MPa	22.9 with $A_c = b_v d$ ACI 318-25M Table 16.4.4.1	
Nominal horizontal shear stress	v_{nh}	2.113	MPa	the mechanism that gives the most	
Maximum tie spacing	s_{max}	600.0	mm	600 mm or 4 x overlay thickness ACI 318-25M 16.4.7.2	
Tie spacing	s / s_{max}	0.500			PASS

L direction - contact width B0

Factored shear at the column face	V_u	728.4	kN	locked-in shear + the old width's share of the increment	
Design horizontal shear strength	ϕV_{nh}	2,266.1	kN	$\phi v_{nh} b_v d$ ACI 318-25M Eq. (16.4.3.1)	
Utilisation	V_u / ϕV_{nh}	0.321			PASS

B direction - contact width L0

Factored shear at the column face	V_u	715.9	kN	locked-in shear + the old width's share of the increment	
Design horizontal shear strength	ϕV_{nh}	3,304.1	kN	$\phi v_{nh} b_v d$ ACI 318-25M Eq. (16.4.3.1)	
Utilisation	V_u / ϕV_{nh}	0.217			PASS

V_u is taken at the column face, the greatest along the contact surface (ACI 318-25M 16.4.2.1, 16.4.3.1): all of the locked-in shear and the old width's share of the increment. Where there are wings, the soil pressure on them can lift an overlay cast with them, so nothing is credited without the minimum ties (16.4.1.2).

10.0 NOT COVERED BY THIS SHEET - CHECK SEPARATELY

- Anchorage of the post-installed dowels and overlay ties in the existing footing: as adhesive anchors (ACI 318-25M Chapter 17), or as post-installed reinforcing bars developed to 25.4.2 and 25.4.9 (17.1.3) with the bar-group breakout of 25.4.11. Shear friction needs f_y developed on both sides of the plane (22.9.5.1), overlay ties likewise (16.4.7.3). ACI 562-25 9.4.4.5 and 9.4.3.4 cap the force credited to post-installed reinforcement at that anchorage value; 9.4.4.6 and 9.4.3.5 do the same on the new-concrete side.

- Development of the existing bottom bars beyond the column-face critical section (ACI 318-25M 13.2.8). They stop at the old footing edge, and the jacket raises the force they must develop.

- Force transfer at the column base: bearing and dowels (ACI 318-25M 16.3 and 22.8).

- Partial contact. The pressure model is linear with the whole base in compression; a case that lifts off is reported as a failed contact check, not solved.

Description	Symbol	Value	Unit	Equation / reference	Status
- Sliding, overturning, settlement and the softer response of the soil under the new wings, which has not been preloaded like the soil under the old footprint. The permissible bearing pressure is an input (ACI 318-25M 13.3.1.1). - Where the bars actually are. The minimum of ACI 318-25M Eq. (8.6.1.2) is checked with the existing bars spread evenly over the old footing and the new bars evenly over the wings; the band of 13.3.3.3, the flexural transfer of the column moment within b_{slab} (8.5.1.1(b)), bar spacing and crack control are not checked. - Seismic requirements for foundations (ACI 318-25M 18.13) and capacity-design demands from the column above. - Bond testing of the prepared surfaces (ACI 562-25 13.4.5), which Tables 9.4.4.1a and 9.4.4.1b call for. - The existing footing during construction: excavation beside it and any loss of bearing or confinement while the jacket is built (ACI 562-25 9.2.1, 9.2.3).					

REFERENCES

- [1] ACI Committee 562, ACI 562-25, Assessment, Repair, and Rehabilitation of Existing Concrete Structures - Code and Commentary, American Concrete Institute, 2025 - 5.3.2 with Table 5.3.2 (strength reduction factors), 9.2 (9.2.1 and R9.2.1 in-place forces, 9.2.3), 9.3.3 (force transfer at interfaces), 9.4 (R9.4.1 load history; 9.4.3.2, 9.4.3.4 and 9.4.3.5 interface tension; 9.4.4 interface shear with 9.4.4.3 roughened substrate, 9.4.4.5, 9.4.4.6 and Tables 9.4.4.1a/b; 9.4.5 combined stresses with 9.4.5.2 and R9.4.5.3), 13.4.5 (bond testing).
- [2] ACI Committee 318, ACI 318-25M, Building Code Requirements for Structural Concrete and Commentary (SI), American Concrete Institute, 2025 - 7.4.3.2 (shear at the face of support), 7.6.1.1 and 8.6.1.1 (minimum flexural reinforcement), Eq. 8.6.1.2 with 8.4.2.2.3 (minimum reinforcement at the column), Eq. 8.4.2.2.2 and 8.4.4.2 with R8.4.4.2.3 (moment transfer in two-way shear), 8.5.1.1, Chapter 13 (13.2.6.2, Table 13.2.7.1, 13.2.7.2, 13.2.8, 13.3.1.1, 13.3.1.2, 13.3.3.3), 16.3 (connections to foundations), 16.4 (16.4.1.2, 16.4.2.1, Eq. 16.4.3.1, Table 16.4.4.1, 16.4.6.1, 16.4.7.2, 16.4.7.3), Chapter 17 with 17.1.3 (anchoring - checked separately), 18.13, Table 20.2.2.4(a) (fy limits), Table 22.2.2.4.3, 22.3.3 (composite Mn), 22.5.3.1, 22.5.4 (composite Vn), Eq. 22.5.5.1.3 (size effect), 22.6 (22.6.2.1, 22.6.3.1, 22.6.4.1, Table 22.6.5.2, 22.6.5.3), 22.8 (bearing), 22.9.4 (Eq. 22.9.4.2, Table 22.9.4.2, Table 22.9.4.4, R22.9.4.5), 22.9.5.1, 25.4.2, 25.4.9 and 25.4.11 (development and bar-group anchorage - checked separately), 26.5.6.2(e).
- [3] ASCE/SEI 41-17, Seismic Evaluation and Retrofit of Existing Buildings, American Society of Civil Engineers, 2017 - C8.7, shallow foundation retrofit, item 2: consideration of the existing contact pressures may be required unless uniform distribution is achieved by shoring or jacking (commentary; this sheet always considers them).
- [4] Antoniou, S., Seismic Retrofit of Existing Reinforced Concrete Buildings, John Wiley & Sons, 2023 - Section 4.18 (foundation strengthening), background only.
- (c) Albert Pamonag Engineering Consultancy. Owner: Albert Pamonag. This calculation sheet is protected; the engineer of record remains responsible for the inputs and for the checks listed as not covered.



Project : Project name

Member : F-1 - Grid A-1

Designed / Date: APEC

03-Oct-2026

THEORETICAL REFERENCE - FOOTING JACKET

ACI 562-25; ACI 318-25M; ASCE 41-17 - clause and printed page under each card

Reference sheet - no inputs. Every cell and picture is locked. Owner: Albert Pamonag - Albert Pamonag Engineering Consultancy.

Equations and provisions behind the calculation sheet

Each card sets out, in this sheet's own notation, the equations a section of the calculation sheet uses; the line under it gives the clause and the printed page where the provision is found. The cards are typeset for this workbook - they are not copies of the standards, which remain the authority.

A. LOAD HISTORY AND BEARING PRESSURE

Unshored: the load on the footing when the jacket is cast stays on the old footprint, and only the increment spreads over the enlarged base. If the load is relieved by shoring or jacking, take $P_0 = M_0 = 0$.

$$q_0 = \frac{P_0}{A_0} \quad q_1 = \frac{P - P_0}{A}$$

$$q_{old, max} = q_0 + q_1 + \frac{6M_0}{B_0 L_0^2} + \frac{6\Delta M L_0}{B L^3} \quad q_{wing, max} = q_1 + \frac{6\Delta M}{B L^2}$$

$$q_{max} \leq q_a \quad q_{min} \geq 0 \text{ (whole base in contact)}$$

The moment terms are added for both directions. In the strength checks the locked-in load stays unfactored and the increment is $P_u - P_0$.

Basis: ACI 562-25 R9.4.1 (p. 85) and R9.2.1 (p. 84); ASCE 41-17 C8.7 item 2 (p. 455); ACI 318-25M 13.3.1.1 (p. 211).

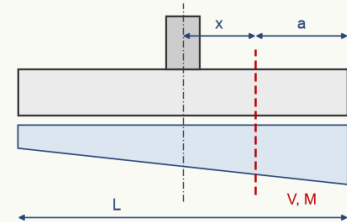
B. PRESSURE BEYOND A SECTION

A footing of length L carries P and M with the whole base in contact. For the part between a section at x from the centre and the edge, with $a = L/2 - x$:

$$V = \frac{Pa}{L} + \frac{6Ma(L/2 + x)}{L^3}$$

$$M_{sec} = \frac{Pa^2}{2L} + \frac{Ma^2(4a + 6x)}{L^3}$$

Locked-in stage: P_0, M_0 on L_0 . Increment: $P_u - P_0, M_u - M_0$ on L . The two are added. Flexure takes x at the column face, one-way shear at d from it, the wing at the old face (increment only).



Basis: Statics. Critical sections: ACI 318-25M Table 13.2.7.1 and 13.2.7.2 (p. 210), 7.4.3.2 (p. 101).

C. FLEXURE OF THE COMPOSITE SECTION

$$a = \frac{\Sigma A_s f_y}{0.85 f'_c b} \quad c = \frac{a}{\beta_1} \quad M_n = \Sigma A_s f_y \left(d - \frac{a}{2} \right)$$

$$\epsilon_t = 0.003 \frac{d_t - c}{c} \quad \phi = 0.90 \text{ if } \epsilon_t \geq \epsilon_{ty} + 0.003, \text{ else } 0.65 + 0.25 \frac{\epsilon_t - \epsilon_{ty}}{0.003} \geq 0.65$$

$$A_{s, min} = 0.0018 A_g \quad d \geq 150 \text{ mm}$$

The entire composite section is used, with the lesser f'_c and no distinction between shored and unshored construction.

Basis: ACI 318-25M 22.3.3.2 to 22.3.3.4 (p. 440), Table 22.2.2.4.3 (p. 439), 7.6.1.1 (p. 102), 8.6.1.1 (p. 122), 13.3.1.2 (p. 212); ACI 562-25 Table 5.3.2 (p. 38).

D. ONE-WAY SHEAR

$$\phi V_c = \phi 0.17 \lambda \sqrt{f'_c} b_w d \quad \phi = 0.75 \quad \sqrt{f'_c} \leq 8.3 \text{ MPa}$$

A rigid footing on soil: no size effect and no reinforcement term. The section is at d from the column face. An overlay not tied to the column is left out of d .

Basis: ACI 318-25M 13.2.6.2(a) (p. 208), 22.5.3.1 (p. 443), 22.5.4.2 to 22.5.4.4 (p. 444); ACI 562-25 Table 5.3.2 (p. 38).

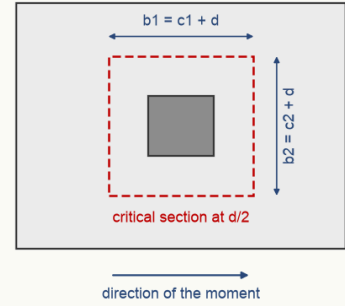
E. TWO-WAY SHEAR WITH MOMENT TRANSFER

$$v_u = \frac{V_u}{b_o d} + \frac{\gamma_v M_{sc} c}{J_c} \quad \gamma_v = 1 - \frac{1}{1 + \frac{2}{3} \sqrt{b_1/b_2}}$$

$$J_c = \frac{d b_1^3}{6} + \frac{b_1 d^3}{6} + \frac{d b_2 b_1^2}{2}$$

$$v_c = \text{least of } 0.33 \lambda \sqrt{f'_c}, \left(0.17 + \frac{0.33}{\beta}\right) \lambda \sqrt{f'_c}, \left(0.17 + \frac{0.083 \alpha_s d}{b_o}\right) \lambda \sqrt{f'_c}$$

Size effect factor 1.0 for a rigid footing on soil. $\alpha_s = 40, 30, 20$ for interior, edge and corner columns. b_1 is the side of the critical section in the direction of the moment.



Basis: ACI 318-25M 22.6.4.1 (p. 451), Table 22.6.5.2 (p. 454), 22.6.5.3 (p. 455), 13.2.6.2(b) (p. 208), Eq. 8.4.2.2.2 (p. 116), 8.4.4.2 and R8.4.4.2.3 (p. 118-119).

F. MINIMUM REINFORCEMENT AT THE COLUMN

$$\text{if } v_{uv} > \phi 0.17 \sqrt{f'_c} \lambda_s \lambda : \quad A_{s, \min} = \frac{5 v_{uv} b_{slab} b_o}{\phi \alpha_s f_y}$$

$$\lambda_s = \sqrt{\frac{2}{1 + d/250}} \leq 1 \quad b_{slab} = c + 3h$$

A lightly reinforced footing punches when the bars near the column yield, below the two-way shear strength. The steel within b_{slab} must supply $A_{s, \min}$ fy.

Basis: ACI 318-25M Eq. 8.6.1.2 (p. 122), 8.4.2.2.3 (p. 116-117), Eq. 22.5.5.1.3 (p. 445).

G. SHEAR FRICTION ON THE OLD FACES

$$V_n = \mu A_v f_y \quad \mu = 1.0 \lambda \text{ (roughened)}, 0.6 \text{ (not roughened)} \quad f_y \leq 420 \text{ MPa}$$

$$V_n \leq \text{least of } 0.2 f'_c A_c, (3.3 + 0.08 f'_c) A_c, 11 A_c \quad (\text{otherwise } 0.2 f'_c A_c, 5.5 A_c)$$

$$V_u = \sqrt{V_{u,w}^2 + (2N)^2} \quad N = \left| \frac{A_{sj} f_{yj}}{2} - 0.85 f'_c a e \right| \frac{M_u}{M_n}$$

N is the part of an adjoining wing's bar force that its own share of the stress block does not balance; it is zero where an overlay is provided. The higher limits need normalweight concrete on a roughened surface. A repair needs an intentionally roughened substrate.

Basis: ACI 318-25M Eq. 22.9.4.2 and Table 22.9.4.2 (p. 472), Table 22.9.4.4 (p. 474), Table 20.2.2.4(a) (p. 414), 22.3.3.1 (p. 440); ACI 562-25 9.4.4.3 (p. 89), R9.4.5.3 (p. 90), Table 5.3.2 (p. 38).

H. HORIZONTAL SHEAR UNDER THE OVERLAY

$$\phi V_{nh} \geq V_u \quad V_{nh} = 0.55 b_v d \text{ (bond, no ties)}$$

$$V_{nh} = \lambda \left(1.8 + 0.6 \frac{A_v f_{yt}}{b_v s} \right) b_v d \leq 3.5 b_v d \text{ (roughened, ties at or above the minimum)}$$

$$A_{vf, \min} = \frac{0.35 b_v s}{f_y} \quad s \leq \text{lesser of } 600 \text{ mm and } 4 t_{top}$$

Where there are wings the soil pressure on them can lift the overlay, so nothing is credited without the minimum ties.

Basis: ACI 318-25M 16.4.1.2 and Eq. 16.4.3.1 (p. 243), Table 16.4.4.1 (p. 244), 16.4.6.1 and 16.4.7.2 (p. 246).

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